

In-Network Normalization of DNP3 Response Timing and Segment Shape on a Programmable Switch

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Abstract—A passive observer of unencrypted DNP3 traffic can fingerprint an outstation from the cross-layer response time and the segment shape of its responses, without decoding the payload. We present an in-network obfuscation layer that runs in the data plane of one Intel Tofino-1 switch at the outstation edge and normalizes both features while preserving DNP3 protocol validity. The defense holds each response in strict-priority switch queues and releases its ACK and response at fixed offsets from the request arrival, so the master-visible cross-layer response time becomes a constant; it then replicates the eligible response and carves it at an existing DNP3 CRC-block boundary into a fixed [28, 21] TCP segment vector, modifying no DNP3 byte and recomputing no CRC. For control commands, a second primitive holds the OPERATE and anchors the master-visible ACK and echo to the request arrival, so the observable gap does not vary with the hidden release delay; the two primitives run on two independent switch scheduling domains. We implement the defense as one P4 program that fits a single Tofino-1 ingress pipe within twelve stages and evaluate it on a physical SEL-751 relay. On the master-facing link the defended response time is a fixed 4.001 ms for read and select in place of the native 1.3 ms and 2.1 ms; each of 1,280 defended responses presents the [28, 21] vector with a CRC-valid, checksum-valid 49-byte reconstruction and no unsplit escape; the control-command gap stays at 4.00 ms across the evaluated release delays; and a read-versus-select timing classifier drops from 0.59 to the 0.50 chance baseline. The evaluation is bounded to one outstation and does not observe the relay-facing link.

Index Terms—DNP3, SCADA, traffic obfuscation, device fingerprinting, programmable switches, industrial control systems.

I. INTRODUCTION

Reconnaissance precedes damage in attacks on power-grid control systems. Before an adversary can open a breaker or forge a measurement that passes the state estimator’s bad-data detection [1], it first learns which devices are on the network and how they behave. In December 2015, remote intruders mapped a Ukrainian distribution utility, then used that knowledge to open breakers and cut power to roughly 225,000 residents [2]. The information a passive observer can extract from unencrypted control traffic is therefore an early and consequential step in the attack chain, not a theoretical concern.

A passive observer does not need to decode a DNP3 payload to fingerprint an outstation. Formby et al. [3] show that cyber-physical devices are identifiable from cross-layer response timing and physical-operation timing, features that survive

the absence of payload decoding. Two wire-visible features carry this signature for DNP3. The cross-layer response time (CLRT) is the interval between the transport-layer ACK and the application-layer response; it reflects the outstation’s internal processing and is stable across sessions. The segment shape is the vector of TCP segment sizes into which the outstation divides a response; because a DNP3 outstation segments deterministically, the shape repeats identically. On our SEL-751 relay, a native read answers with a single 49-byte segment and a CLRT median of 1.3 ms, while a control select answers at 2.1 ms, so the two transaction classes are separable by timing alone. Prior traffic-analysis work shows that such size-and-timing side channels recover content even on encrypted sessions [4].

Normalizing these features in the network is desirable for two reasons. *First*, an in-network defense needs no change to the master or the vendor-supplied outstation, which a legacy substation cannot modify. *Second*, it suppresses the reconnaissance signal before an attack executes, rather than detecting the attack after the fact. The deployment is, however, constrained on both sides. *First*, transport encryption is usually absent on legacy DNP3 and cannot be assumed, and encryption alone would not hide the timing and segment-count features that survive tunneling [4]. *Second*, the outstation firmware is fixed and vendor-supplied, so a defense cannot change how the device builds its frames, and any modification of the response bytes breaks the per-block CRCs the master validates [5]. *Third*, generic traffic-analysis defenses do not fit: per-packet padding leaves exploitable size and count features [6], morphing one class into another assumes control of the sending endpoint [7], and website-fingerprinting defenses such as adaptive padding target browser traffic over an anonymity network with host cooperation [8]. *Last*, host-based delay or a controller-mediated release changes the endpoints or adds a control-plane fast path that a legacy substation deployment cannot accommodate.

There exists a gap for a defense that normalizes both the response timing and the eligible segment shape of DNP3 traffic in the network while preserving protocol validity and requiring no endpoint change. We address this gap with an in-network obfuscation layer that runs entirely in the data plane of one Intel Tofino-1 switch placed at the outstation edge. The defense holds each response in strict-priority switch queues

and releases its ACK and response at fixed policy offsets from the request arrival, so the master-visible CLRT becomes a constant. It then replicates the eligible 49-byte response and carves it at an existing DNP3 CRC-block boundary into a fixed [28,21] segment vector, so no DNP3 byte is modified and the master reassembles the identical response. We refer to this timing-and-segmentation path as response shaping.

The defense exposes selectable modes at runtime. A passthrough mode forwards traffic natively. Timing modes hold the ACK, the response, or both to their policy offsets; the combined mode, which holds both, is the one that normalizes the CLRT and drives the response-shaping path above. A separate mode protects control commands.

The control-command mode is necessary because the Select-before-Operate exchange exposes an operation-timing fingerprint: the delay from an OPERATE to its confirmation reflects the relay’s actuation. Response shaping alone cannot hide this delay, because releasing the OPERATE on a fixed schedule would still reveal it. The control-command mode blocks the OPERATE and is designed to release it once at a secret internal delay while anchoring the master-visible ACK and echo to the original arrival time, so the observable gap does not vary with the hidden delay. Response shaping and the control-command mode cannot share one strict-priority queue ladder, so the defense runs them on two independent switch scheduling domains, both internal to the same Tofino and the same program; Section III explains the scheduling conflict that forces the split.

We evaluate the defense on a testbed with a physical master, one Tofino-1, and a physical SEL-751 relay, comparing native and defended traffic captured on the same binary under a runtime mode change. On the master-facing link, the defended CLRT is a fixed 4.001 ms (standard deviation 0.02 ms) for both read and select, in place of the native 1.3 ms and 2.1 ms; every one of 1,280 defended eligible responses presents the [28,21] segment vector with a sequence-contiguous, CRC-valid, checksum-valid 49-byte reconstruction and no unsplit escape. The control-command gap stays at 4.00 ms across the evaluated release delays, and a classifier that separated read from select by CLRT with 0.59 balanced accuracy on native traffic drops to the 0.50 chance baseline on defended traffic. The guarded control test permitted only isolated output points and refused a breaker-close index, and all 32 relay outputs remained open throughout.

The evaluation is bounded. We test one outstation, so the defense replaces one device’s CLRT and segment-shape signature with a policy rather than establishing indistinguishability among devices. The relay-facing link was not captured, so the single release at the secret delay is verified by the implementation model but not observed on the wire, and we do not claim exactly-once delivery to the relay. The defense normalizes the observed segment vector, not the total 49-byte payload length.

To the best of our knowledge, this is the first in-network defense to normalize both the response timing and the segment shape of a live industrial-control protocol while preserving its

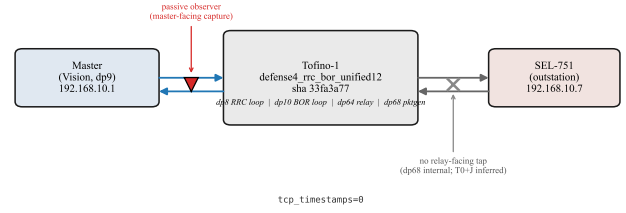


Fig. 1. Testbed and observation model. The master reaches the physical SEL-751 through one Tofino-1 switch. The passive adversary observes only the master-facing link; the relay-facing link and the switch-internal loopbacks (dp8 for response shaping, dp10 for the control-command hold) are not observed.

bytes and CRCs. This paper makes the following contributions.

- **A hardware timing primitive for absolute-offset response shaping.** The response-shaping path holds the ACK and response in strict-priority reservoirs and releases them at fixed offsets from the request timestamp, normalizing the master-visible CLRT to a constant, with selectable timing modes.
- **Deterministic in-network segment-shape normalization.** We use packet replication and egress carving to emit an eligible response as a fixed [28, 21] segment vector on an existing DNP3 CRC-block boundary, preserving every DNP3 byte and CRC.
- **A control-command timing mode.** The control-command mode blocks the OPERATE and anchors the master-visible ACK and echo to the request arrival, and we identify why the two mechanisms require two independent switch scheduling domains.
- **A one-program implementation and a bounded hardware evaluation.** We implement the defense as one P4 program that fits a single Tofino-1 ingress pipe within twelve stages, and we evaluate it on a physical SEL-751 with a reproducible capture-to-claim analysis, an explicit safety result, and a stated evidence boundary.

II. THREAT MODEL, ASSUMPTIONS, AND GOALS

A. System and observation model

Figure 1 shows the setup. We consider a DNP3-over-TCP link between one master and one outstation, the SEL-751 relay, with a Tofino switch on the path at the outstation edge. The master polls the outstation with reads and, occasionally, issues a Select-before-Operate control. The adversary observes the master-facing link, that is, the segment between the master and the switch, and records TCP/IP headers, TCP segment boundaries, payload lengths, and arrival times. The adversary does not observe the relay-facing link or the switch-internal loopbacks. We assume the traffic is unencrypted, as legacy DNP3 usually runs without transport security across shared substation and utility networks, so read access to the link is within reach of an adversary that has already gained a foothold.

B. Adversary capabilities

Passive fingerprinting. For every claim in this paper the adversary is passive: it captures traffic from a mirrored port or a tap and neither modifies, injects, nor blocks a packet. Following Formby et al. [3], the adversary fingerprints a device from features that survive the absence of payload decoding. Two features are in scope. The cross-layer response time is the interval between the transport-layer ACK and the application-layer response; it reflects the outstation's internal processing and is stable across sessions. The segment shape is the vector of TCP segment sizes into which a response is divided, which a deterministic outstation produces identically each time. The adversary knows the protocol and may know the defense policy; its advantage comes from observing these features, not from a secret.

Out of scope for the threat. An adversary that observes the relay-facing link, that breaks a protected channel, or that actively manipulates traffic is out of scope. Passive observation of the master-facing link is the honest baseline for unencrypted DNP3, and size and timing cues survive some tunneling regardless.

C. Trusted computing base and deployment assumptions

The master, the switch and its loaded program, and the outstation are trusted. The defense runs entirely in the switch data plane; there is no host agent, controller fast-path, or endpoint change. We assume no TCP timestamp option on the protected flow, because the option is an independent timing channel that would expose the held delay; the pipeline gates on this option and the evaluated captures carry none. Shaping applies to the eligible response class, here the 49-byte read and control responses; other traffic forwards natively.

D. Security goals

We state the security goals (G1–G4, G6) as testable properties of the master-facing observation; G5 is an evaluation-validity property that lets the native and defended comparison isolate the defense.

- **G1 (CLRT normalization).** The defended cross-layer response time is a fixed policy value independent of the transaction, so $CLRT = R - A$ replaces the outstation's native timing.
- **G2 (fixed segment vector).** Every eligible response presents the same TCP segment vector [28, 21] in place of the native single segment.
- **G3 (protocol-valid reconstruction).** The master re-assembles a sequence-contiguous, DNP3-block-CRC-valid, IP/TCP-checksum-valid response.
- **G4 (master-visible J independence).** For a control command, the master-visible ACK and echo placement does not vary with the internal release delay J .
- **G5 (testbed preservation).** The defense changes only a runtime mode; the native and defended comparison isolates the mechanism.
- **G6 (safety).** The defense actuates no physical output.

E. Non-goals and unverified properties

The defense does not aim for the following, and the evidence does not establish them.

- **Relay-facing verification.** The relay-facing release time $T_0 + J$ and the number of copies that reach the relay are not observed; the design intends a single release, but the captures establish only that the master issued one OPERATE per transaction.
- **Full length hiding.** The reassembled response remains 49 bytes; the defense normalizes the observed segment vector, not the total payload length, and a full TCP reassembler recovers the 49 bytes.
- **Byte identity.** No paired source-side oracle was captured, so we do not claim byte identity between an emitted response and a source frame.
- **Multi-device anonymity.** We evaluate one SEL-751; the mechanism replaces that device's native CLRT and segment-shape signature with a policy, and does not establish indistinguishability among devices or resistance to every fingerprinting feature. The evaluated classifier separates READ from SELECT by CLRT, a transaction-class task, not a device-identity task.
- **Encrypted-payload semantics and active adversaries.** The defense preserves protocol validity of an unencrypted payload; it neither adds confidentiality nor defends against traffic manipulation.

III. IMPLEMENTATION

To evaluate in-network obfuscation on production-representative hardware, we implemented the defense as a single P4₁₆ program for one Intel Tofino-1 switch, together with a control-plane configuration layer and a guarded control-test harness. The data plane comprises six major parts: (i) a parser and transaction classifier; (ii) an absolute-timestamp deadline stage; (iii) a packet generator that seeds strict-priority blocker reservoirs; (iv) the response-shaping release, replication, and egress carving path; (v) the control-command epoch-and-hold state machine; and (vi) two independent Traffic-Manager scheduling domains. A single terminal table applies every port, queue, drop, bypass, and multicast effect. The program fits in one ingress pipe within the twelve-stage match-action budget.

A. Physical topology and port roles

The master, the switch, and the outstation sit in one line: the master reaches the switch on device port dp9, and the switch reaches the physical SEL-751 on dp64. Three ports are internal to the switch and carry no external cable: dp8 recirculates responses for the response-shaping ladder, dp10 holds the OPERATE for the control-command hold, and dp68 injects generated tokens and clones. The adversary's observation point is the master-facing link at dp9; the relay-facing link at dp64 and the internal ports are not part of the observed traffic. We treat dp8, dp10, and dp68 as switch implementation details, not additional inline devices.

TOFINO-1 · ONE PIPE · 12 INGRESS MAU STAGES · TM STRICT-PRIORITY QUEUES
Unified RRC + BOR pipeline and Traffic Manager

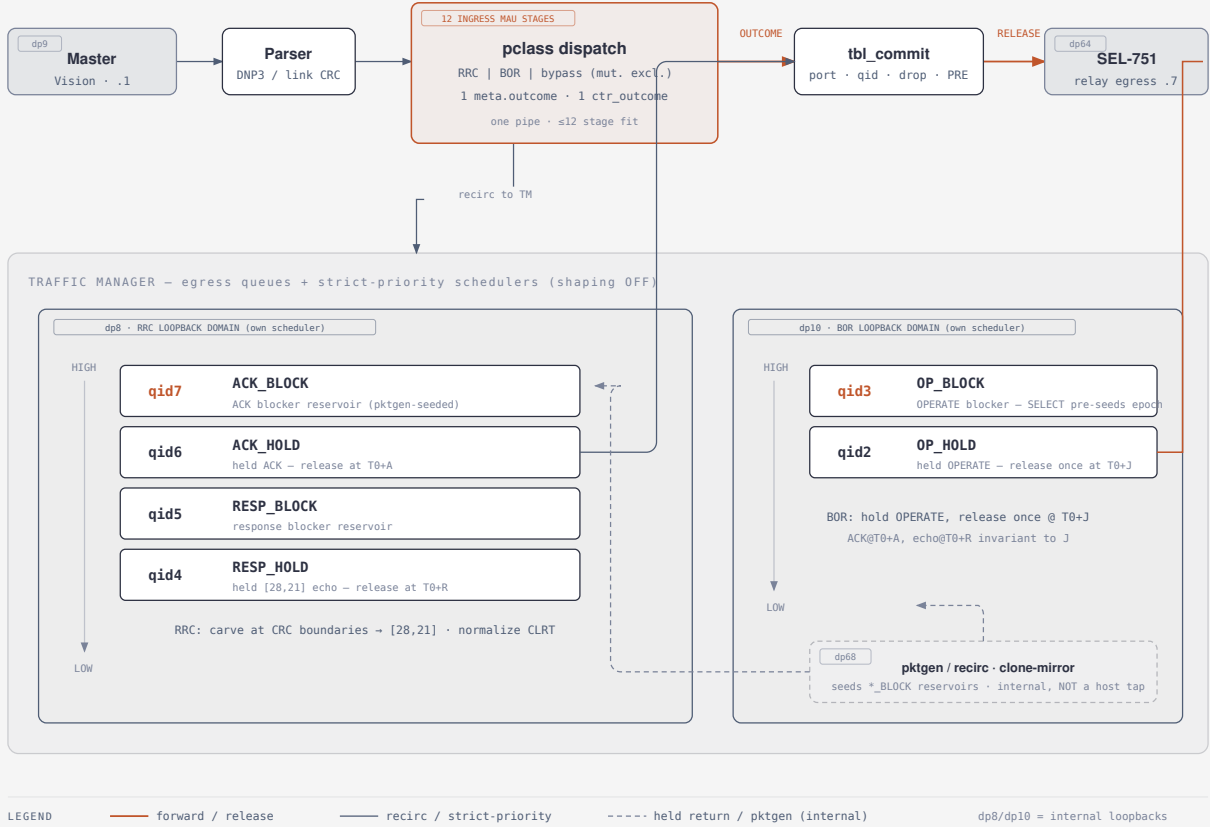


Fig. 2. One-program pipeline and the two scheduling domains. The parser and match-action stages compute a single meta.outcome that a terminal tbl_commit applies. The Traffic Manager holds the response-shaping reservoirs on dp8 (queues qid7–qid4) and the control-command queue pair on dp10 (qid3,qid2); the packet generator on dp68 seeds the blocker reservoirs. Egress carves the eligible response to [28, 21].

B. One-program pipeline and the single commit

One outcome, one commit. Figure 2 shows the pipeline. Earlier match-action logic computes a compact result meta.outcome; a terminal table tbl_commit keyed on that result applies the port, queue, drop, bypass, or multicast effect and is the only stage that writes those fields. The map is constant and total over every defined outcome, and its default action forwards the packet unmodified rather than dropping it, so a program loaded before the control plane runs still forwards traffic. This structure keeps the disposition auditable from a single table and is what lets the two mechanisms share one ingress pipe within twelve stages: a decision-table flatten collapses the branch tree into two mutually exclusive ternary tables that co-place in one stage and feed the single commit.

C. Parser and transaction classification

The parser recognizes Ethernet, IPv4, TCP, and the DNP3 link, transport, and application headers, and rejects IP fragments and non-TCP traffic to the forwarding default. Classifi-

cation folds READ, SELECT, and OPERATE into one role at parse time and re-derives the DNP3 function code inside the pipeline, producing a control byte that gates every downstream decision. A returning recirculated packet is identified by its ingress port and then keyed on its content, not on the port, so the same logic applies regardless of which internal loopback delivered it. Generated tokens and clones are recognized by a parser value set and a leading tag byte.

D. Absolute timestamps and deadlines

Anchoring to the request. On the request or OPERATE, the pipeline records an ingress timestamp T_0 and arms two deadlines, T_0+A for the ACK and T_0+R for the response, from the policy offsets A and R . Anchoring to T_0 rather than to the relay’s later ACK is required for the control-command timing argument: because the ACK exists only after the held OPERATE is released, an ACK-relative deadline would expose the hidden hold. The later relay ACK cannot re-anchor a deadline once T_0 has set it. The offline model verifies this

anchoring against a mutant that arms the deadline at the ACK, which the invariant kills.

E. Packet generation and blocker reservoirs

Reservoirs as timing gates. A held ACK or response sits in its queue behind a higher-priority reservoir of generated blocker packets; strict priority drains the blockers first, so the held packet cannot leave until its reservoir empties at the policy deadline. The on-chip packet generator on dp68 fills these reservoirs; it never generates master or relay traffic. Two generator profiles are pinned to distinct tag bytes so that a READ or SELECT arm and an OPERATE hold trigger different reservoir sets without ambiguity, a separation that a single shared tag did not provide. The reservoir depth is fixed at $K=64$.

F. Response shaping: release, replication, and carving

Response shaping normalizes the segment shape and the ACK-to-response timing of an eligible 49-byte response in four steps. *First*, the strict-priority reservoirs hold the real ACK and the real response on the dp8 ladder across queues `qid7` (ACK blocker), `qid6` (ACK hold), `qid5` (response blocker), and `qid4` (response hold). *Second*, the ACK and the response release at the absolute offsets T_0+A and T_0+R , so the master-visible cross-layer response time $CLRT = R - A$ is a fixed policy value. *Third*, the release event steers the eligible response into a Packet-Replication-Engine multicast group, which produces two master-facing copies. *Last*, egress carving emits the 28-byte prefix from one copy and the 21-byte suffix from the other, advancing the suffix TCP sequence number by 28 so the two segments abut. The physical relay produces the response; PRE replication produces the two copies used for segmentation, and no clone or mirror fabricates the response. No DNP3 byte is modified and no DNP3 CRC is recomputed; only the IPv4 and TCP checksums are recomputed over each emitted window. Byte 28 is a selected existing DNP3 CRC-block boundary that simplifies deterministic carving and validation; TCP may segment a byte stream at any offset, so this is a design choice, not a protocol requirement, and the reassembled response remains 49 bytes. Figure 3 shows the 49-byte response and the carve: the prefix is the 10-byte link header plus the first CRC-protected data block, and the suffix is the remaining blocks, so each emitted segment ends on an already-valid CRC.

G. Control commands: block the OPERATE, then release it once

The control-command hold governs the one control command. **Preparing an epoch.** A SELECT crossing the pipeline pre-seeds the OPERATE blocker reservoir on dp10, so the hold is armed before the OPERATE arrives; the epoch is an internal per-transaction marker, not the DNP3 sequence. **Holding and releasing.** The switch admits one matching OPERATE, holds it in `qid2` behind the `qid3` reservoir, and is designed to release it once at T_0+J , where J is drawn in the data plane from a bounded codebook so that a passive

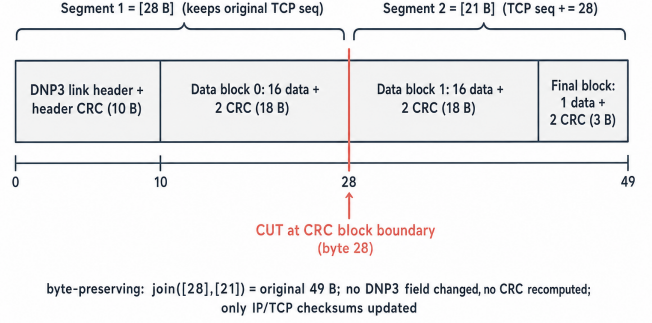


Fig. 3. The 49-byte SEL-751 response and the [28, 21] carve. The cut at byte 28 falls on an existing DNP3 CRC-block boundary; the prefix keeps the original TCP sequence number and the suffix advances it by 28, so the master reassembles the identical 49-byte response. No DNP3 byte is modified and no CRC is recomputed.

observer cannot predict it from public header fields. **Master-visible placement.** The ACK and echo release to the master at T_0+A and T_0+R , anchored to the original T_0 and independent of J , so the observable gap $\text{echo} - \text{ACK} = R - A$ carries no information about the hold. **Retirement.** After release the generation is kept as a spent marker until the next SELECT; a retransmitted OPERATE that reads the spent marker is classified a duplicate and dropped. The hardware captures establish the master-visible placement and the invariant gap; the offline model verifies the release-at- T_0+J state machine; the relay-facing T_0+J timestamp and delivery multiplicity are not directly observed, because dp68 is internal and no relay-facing tap was captured.

H. Two independent scheduling domains

Why two domains. Response shaping seeds ACK and response reservoirs on high-priority queues. On one shared strict-priority ladder those reservoirs starve the OPERATE blocker and hold queues after the OPERATE arrives, which shifts the release toward A or R instead of T_0+J . The final design places the response-shaping ladder on dp8 and the control-command queue pair on dp10, so each mechanism arbitrates on its own egress scheduler. Both loopbacks are internal paths in the same physical Tofino and the same P4 program; the separation removes the coupling without adding an inline device or a second pipe.

I. Control plane and readback

The control-plane layer brings up the ports, creates the Traffic-Manager queues at descending strict-priority levels with shaping disabled, installs the multicast group and mirror session, writes the session and timing parameters and the J codebook, and initializes the control-command registers. Every write is read back and compared, and a mismatch or a configuration warning leaves the mechanism disabled rather than partially configured. There is no configuration snapshot;

rollback is a forward, readback-checked teardown to a plain forwarding state.

J. Failure behavior

Fail open. On a missing state, an unarmed deadline, or an unrecognized class, the terminal commit forwards the packet unmodified; the design never drops on the default path. A retransmitted OPERATE after release is dropped as a duplicate rather than re-forwarded. Traffic outside the eligible 49-byte class, unsupported DNP3 functions, and malformed frames bypass shaping and forward natively. The control-command timing argument holds only on a flow without TCP timestamps; the pipeline gates on the timestamp option and the evaluated captures carry none.

K. Safety guard and testbed restoration

The control-test harness that issues a physical SELECT and OPERATE is guarded: it permits only the isolated CROB points $\{1, 3\}$ and refuses the breaker-close index 6. Across every guarded transaction, all 32 relay outputs remained OPEN, and no breaker operation was attempted. Native and defended trials use the same master, SEL-751, links, switch, capture clock, and loaded binary, separated only by a runtime mode change, so the comparison isolates the defense rather than the environment.

L. Resource fit and implementation limitations

The program compiles for one Tofino-1 ingress pipe within the twelve-stage budget; the control-command state uses four registers in a domain gated so that an OPERATE token never reaches the response-shaping registers. The implementation does not observe the relay-facing link, does not normalize the total reassembled response length, and is evaluated on one outstation; Section IV reports the measured consequences of these bounds. **Repeatability.** The loaded source and binary are identified by hash, the analysis regenerates deterministically from the captured traces, and a manifest verifies every derived artifact.

IV. EVALUATION

We evaluate the defense on a testbed with a physical master, one Tofino-1, and a physical SEL-751 relay. Native and defended traffic are captured on the same loaded binary, separated only by a runtime mode change, so the comparison isolates the defense. The defended captures carry no TCP timestamp option. The analysis regenerates deterministically from the traces, and a manifest verifies every derived artifact. **Timing.** The native CLRT median is 1.272 ms for read (mean 1.880, standard deviation 1.354, $n=999$) and 2.107 ms for select ($n=488$), with heavy tails. The defended CLRT median is 4.001 ms for both classes, with standard deviation 0.022 ms (read) and 0.021 ms (select), matching the policy $R - A = 4$ ms; the master-facing offsets of about 21 and 25 ms include a documented path and capture offset of roughly 1 ms over the configured 20 and 24 ms. The defense adds 2.73 ms to a read and 1.89 ms to a select.

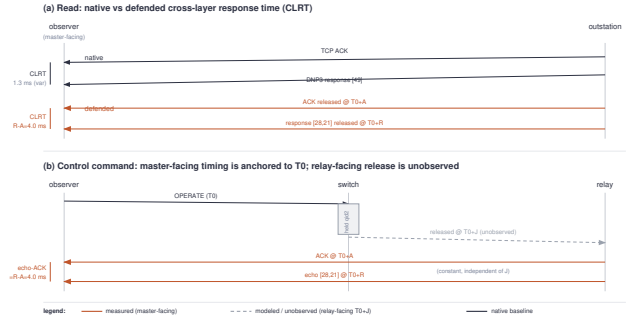


Fig. 4. Master-facing timing of a control command. The ACK and echo are placed at T_0+A and T_0+R , anchored to the OPERATE arrival T_0 , so the observable gap $R - A$ is fixed. The internal release at T_0+J is on the relay-facing link and is not captured (dashed), so it is verified by the implementation model rather than measured.

Segment shape and reconstruction. The native read is a single 49-byte segment. All 1,280 defended eligible responses (600 read, 500 select, and 90 select and 90 operate from the control transactions) present the [28, 21] segment vector. Each reconstruction is sequence-contiguous, DNP3-block-CRC-valid, and IP/TCP-checksum-valid, and no defended response escapes as an unsplit 49-byte segment. Because no paired source-side capture exists, we report a CRC-valid 49-byte reconstruction rather than byte identity to a source frame.

Control-command timing. Figure 4 shows the master-facing ACK and echo placement for a control command. The echo-minus-ACK gap is 4.00 ms and does not vary with the internal release delay: across $J \in \{2, 6, 12\}$ ms the pooled gap median is 4.002 ms with standard deviation 0.026 ms ($n=90$).

The relay-facing release time and the number of copies delivered to the relay are not captured; the offline model verifies the single release at T_0+J , and the captures establish that the master issued one OPERATE per transaction.

Fingerprint suppression. We quantify the leak with mutual information between the transaction class and the CLRT, using common bins and a permutation null. The native mutual information is 0.424 bits, well above the null band, and the defended value is 0.0018 bits, inside it. A logistic classifier trained to separate read from select by CLRT reaches 0.592 balanced accuracy on native traffic and 0.500, the chance baseline, on defended traffic. This is a transaction-class task on one device, not a device-identity task; the defense replaces the evaluated CLRT and segment-shape features for the SEL-751 rather than establishing indistinguishability among devices.

Safety and resource fit. The guarded control test permitted only the isolated CROB points $\{1, 3\}$ and refused the breaker-close index 6; all 32 relay outputs remained OPEN throughout, and no breaker operation was attempted. The program compiles for one Tofino-1 ingress pipe within the twelve-stage budget.

V. DISCUSSION

The defense replaces two master-facing features of one outstation with a fixed policy and preserves the protocol. It

does not hide the total 49-byte payload length, which a full TCP reassembler recovers, and it does not establish multi-device anonymity or resistance to every traffic-analysis feature. Extending the eligible class beyond the 49-byte responses would require a segment-vector policy for each response size, which the CRC-block structure supports but we have not evaluated. The strongest open item is relay-facing verification: a tap on the relay-facing link would turn the modeled single-release property into a measured one, and we leave this instrumentation to future work.

VI. CONCLUSION

We presented the defense, an in-network defense that normalizes the cross-layer response time and the TCP segment shape of DNP3 traffic in the data plane of one programmable switch, without modifying a DNP3 byte or recomputing a CRC. A timing primitive releases the ACK and response at fixed offsets from the request, a replication-and-carving path fixes the eligible segment vector at an existing CRC boundary, and a control-command primitive hides operation timing while anchoring the master-visible acknowledgment to the request. On a physical SEL-751 relay, the defense normalizes the defended CLRT to 4.001 ms, presents every eligible response as [28, 21] with a CRC-valid reconstruction, holds the control-command gap at 4.00 ms across release delays, and drives a read-versus-select timing classifier to chance, within a stated single-device and master-facing evaluation boundary.

VII. RELATED WORK

Device fingerprinting in cyber-physical systems. Formby et al. [3] fingerprint ICS devices from cross-layer response timing and physical-operation timing, distinguishing device types and instances without decoding payloads, and GTID [9] identifies devices and device types from wire-side timing signatures. Physical-layer signatures fingerprint devices even more directly, from clock skew observed remotely [10] to the process-actuator physics of a cyber-physical device [11]. This line establishes the threat the defense addresses: observable timing and shape features identify an outstation. It is an attacker capability rather than a defense. The defense differs in objective; we suppress the master-facing CLRT and segment-shape features these techniques exploit, for one evaluated device, rather than proposing a new fingerprint.

Network-timing traffic analysis and its defenses. Inter-event timing alone leaks secret content, from keystroke intervals on SSH [12] to modern website-fingerprinting attacks that recover a browsed page from packet sizes and timing with a deep classifier [4], although such accuracy is contested at realistic open-world scale [13]. The corresponding defenses shape the traffic: adaptive padding inserts dummy packets to break the inter-packet timing pattern [8], and comprehensive side-channel mitigation shapes a tenant’s traffic to a schedule that leaks no secret [14]. These defenses target browser traffic over an anonymity network, or cloud tenants, and assume a cooperating end host or hypervisor that can add cover traffic. Unlike these host-cooperative defenses, our mechanism

normalizes the ACK-to-response timing of a protocol-bound DNP3 exchange in the network, with no endpoint change and no added application bytes, and it fixes a specific cross-layer interval rather than obscuring an inter-packet distribution.

Packet-size padding, morphing, and segmentation. Size-based countermeasures reshape the length signature of a flow. Traffic morphing optimally transforms one class’s packet-size distribution into another’s [7], and later analysis shows that per-packet padding and fragmentation leave count and size features that a classifier still exploits [6]. Constant-rate padding such as Tamaraw bounds leakage by fixing the packet size and interval, at a cost acceptable for one flow but prohibitive across a device fleet [15], while burst molding collides distinct activities onto one pattern with lower overhead [16]. The same size-and-rate channel identifies devices even under encryption, where traffic shaping is the standard countermeasure [17], and shaping can be made differentially private to bound the residual leak [18]. These lines change or pad bytes and target the end host. The defense differs in mechanism; it neither pads nor rewrites DNP3 bytes, but re-segments the response only at existing CRC-block boundaries, so the reassembled message and every per-block CRC are preserved while the observed segment vector is fixed.

Anti-reconnaissance and moving-target defense for power grids. A body of work disrupts grid reconnaissance rather than detecting its use. DefRec virtualizes physical functions to present deceptive content and connectivity to an attacker [19], RAINCOAT randomizes data acquisition and injects decoy measurements to mislead attack strategy [20], and a companion testbed evaluates such anti-reconnaissance approaches on grid cyber-physical infrastructure [21]. A parallel detection line adapts specification-based intrusion detection to DNP3 [22]. In-network obfuscation has denied reconnaissance before: NetHide rewrites data-plane topology to blunt link-flooding attacks while keeping path tracing usable [23]. These defenses operate at the content, connectivity, topology, or detection layer. Our defense instead operates at a different observation point: it obfuscates the wire-visible size and timing of the transport exchange itself, in the network and below the application semantics those defenses reshape, and it complements rather than replaces them.

Timing and shaping on programmable switches. Programmable data planes make in-network traffic shaping practical. The P4 model exposes a programmable parser and match-action pipeline [24], and push-in-first-out queues enable programmable packet scheduling at line rate [25]. PIFO is a hardware abstraction; SP-PIFO shows that rank-ordered scheduling can be approximated with a few strict-priority queues on today’s switches [26], the primitive class our deadline-driven response scheduler is built on. Ditto pads packets and injects chaff on a programmable switch to make WAN traffic follow a deterministic pattern independent of the payload, at line rate and without end-host changes [27]. Ditto is the closest in-network precedent. It adds bytes and packets to reach a fixed volume pattern, which is appropriate for opaque WAN traffic but not for a protocol whose bytes and CRCs the receiver

validates. Recent systems bring traffic-analysis defense fully into the programmable data plane: Securitas obfuscates flows against traffic analysis with flexible in-network shaping [28], Minos mounts a lightweight, dynamic defense against traffic analysis on programmable switches [29], and, closest to our setting, Hu and Lin enhance industrial network-protocol security with programmable switches [30]. The defense differs in two ways from all of these: it is byte-preserving and carves only at existing DNP3 boundaries, so the outstation's protocol remains valid, and it adds a control-command mode that hides operation timing without exposing the hold to a passive observer.

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